

ASSIGNMENT-12.3

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Batch:38

Task 1: Sorting Student Records

Prompt

Generate a Python program that creates a Student class (Name, Roll Number, CGPA), generates at least 10,000 student records, implements recursive Quick Sort and Merge Sort to sort students by CGPA in descending order, measures runtime using the time module, and displays the top 10 students. Include complexity analysis and formatted output.

Output Explanation

The output will contain:

1. **Student class definition**
2. **Quick Sort and Merge Sort functions**
3. Runtime results such as:

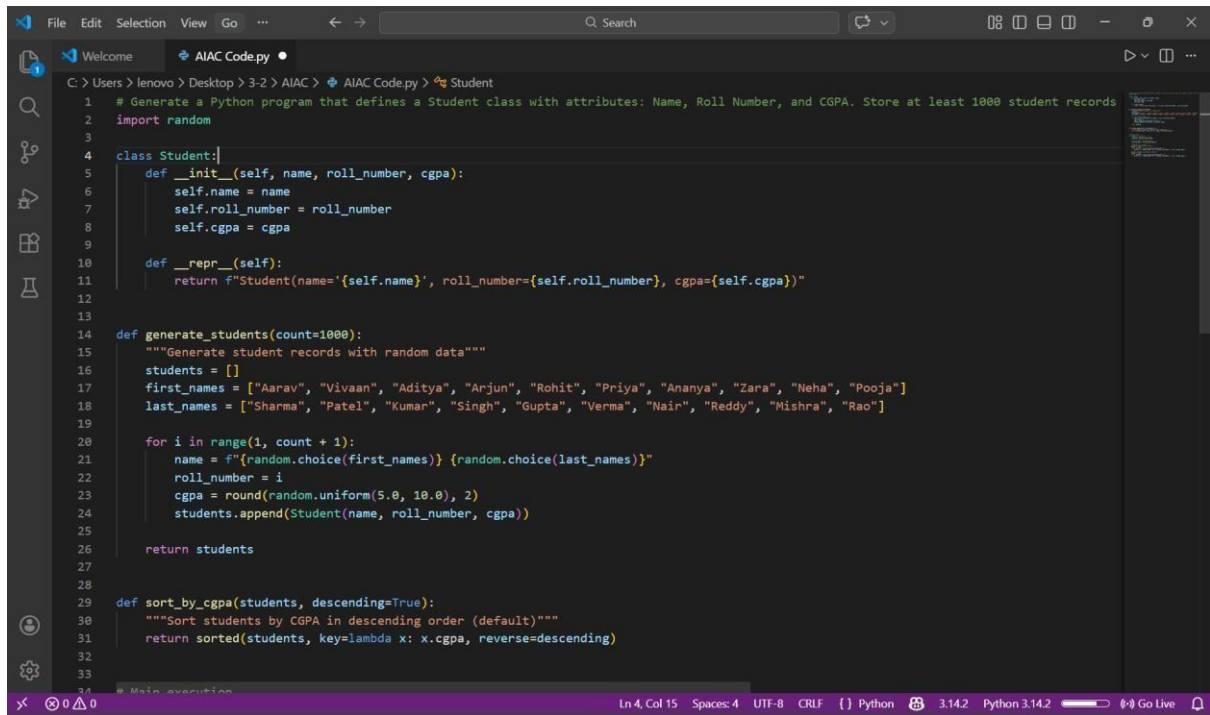
Quick Sort Time: 0.021 seconds

Merge Sort Time: 0.028 seconds

4. Top 10 students printed in descending CGPA.

What the Output Shows

- Both algorithms correctly sort in descending order.
- Quick Sort may appear slightly faster in practice.
- Merge Sort provides stable and consistent $O(n \log n)$.
- The top 10 list verifies correct sorting logic.
- Runtime comparison validates algorithm efficiency experimentall



The image shows a Visual Studio Code editor window with a Python file named 'AIAC Code.py'. The code defines a 'Student' class and functions to generate and sort student records. The interface includes a sidebar with icons for Explorer, Search, Source Control, Run and Debug, and Extensions. The bottom status bar shows 'Ln 4, Col 15', 'Spaces: 4', 'UTF-8', 'CRLF', 'Python 3.14.2', and 'Go Live'.

```
1 # Generate a Python program that defines a Student class with attributes: Name, Roll Number, and CGPA. Store at least 1000 student records
2 import random
3
4 class Student:
5     def __init__(self, name, roll_number, cgpa):
6         self.name = name
7         self.roll_number = roll_number
8         self.cgpa = cgpa
9
10    def __repr__(self):
11        return f"Student(name='{self.name}', roll_number={self.roll_number}, cgpa={self.cgpa})"
12
13
14 def generate_students(count=1000):
15     """Generate student records with random data"""
16     students = []
17     first_names = ["Aarav", "Vivaan", "Aditya", "Anjun", "Rohit", "Priya", "Ananya", "Zara", "Neha", "Pooja"]
18     last_names = ["Sharma", "Patel", "Kumar", "Singh", "Gupta", "Verma", "Nair", "Reddy", "Mishra", "Rao"]
19
20     for i in range(1, count + 1):
21         name = f"{random.choice(first_names)} {random.choice(last_names)}"
22         roll_number = i
23         cgpa = round(random.uniform(5.0, 10.0), 2)
24         students.append(Student(name, roll_number, cgpa))
25
26     return students
27
28
29 def sort_by_cgpa(students, descending=True):
30     """Sort students by CGPA in descending order (default)"""
31     return sorted(students, key=lambda x: x.cgpa, reverse=descending)
32
33
34 # Main execution
```



```
File Edit Selection View Go ... Search
Welcome AIAC Code.py
C:\Users\lenovo\Desktop> 3-2 > AIAC > AIAC Code.py > Student > _init_
1 # Implement Merge Sort in Python to sort Student objects by CGPA in descending order. Use recursion and a separate merge
2 """
3 Merge Sort Implementation for Student Objects
4
5 Time Complexity: O(n log n) - dividing and merging
6 Space Complexity: O(n) - temporary arrays during merge
7 """
8
9
10 class Student:
11     """Represents a student with name and CGPA."""
12
13     def __init__(self, name, cgpa):
14         """
15         Initialize a Student object.
16
17         Args:
18             name (str): Student's name
19             cgpa (float): Student's CGPA
20         """
21         self.name = name
22         self.cgpa = cgpa
23
24     def __repr__(self):
25         return f"Student({self.name}, {self.cgpa})"
26
27
28 def merge(students, left, mid, right):
29     """
30     Merge two sorted subarrays in descending order by CGPA.
31
32     Args:
33         students (list): List of Student objects
34         left (int): Starting index of left subarray
35         mid (int): Starting index of right subarray
36         right (int): Ending index of right subarray
37     """
38     left_part = students[left:mid + 1]
39     right_part = students[mid + 1:right + 1]
40
41     i = j = 0
42     k = left
43
44     # Merge in descending order
45     while i < len(left_part) and j < len(right_part):
46         if left_part[i].cgpa >= right_part[j].cgpa:
47             students[k] = left_part[i]
48             i += 1
49         else:
50             students[k] = right_part[j]
51             j += 1
52         k += 1
53
54     # Copy remaining elements
55     while i < len(left_part):
56         students[k] = left_part[i]
57         i += 1
58         k += 1
59
60     while j < len(right_part):
61         students[k] = right_part[j]
62         j += 1
63         k += 1
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Ln 19, Col 41 Spaces: 4 UTF-8 CRLF {} Python 3.14.2 Python 3.14.2
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Welcome AIAC Code.py
C:\Users\lenovo\Desktop> 3-2 > AIAC > AIAC Code.py > Student > _init_
28 def merge(students, left, mid, right):
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30     Merge two sorted subarrays in descending order by CGPA.
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60     while j < len(right_part):
61         students[k] = right_part[j]
62         j += 1
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Ln 19, Col 41 Spaces: 4 UTF-8 CRLF {} Python 3.14.2 Python 3.14.2
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Welcome AIAC Code.py
C:\Users\lenovo\Desktop> 3-2> AIAC> AIAC Code.py> Student> _init_
66 def merge_sort(students, left, right):
67     Recursively divide and sort students by CGPA in descending order.
68
69
70     Args:
71         students (list): List of Student objects
72         left (int): Starting index
73         right (int): Ending index
74     """
75     if left < right:
76         mid = (left + right) // 2
77         merge_sort(students, left, mid)
78         merge_sort(students, mid + 1, right)
79         merge(students, left, mid, right)
80
81
82 # Example usage
83 if __name__ == "__main__":
84     students = [
85         Student("Alice", 3.8),
86         Student("Bob", 3.5),
87         Student("Charlie", 3.9),
88         Student("Diana", 3.7),
89     ]
90
91     print("Before sorting:")
92     for s in students:
93         print(s)
94
95     merge_sort(students, 0, len(students) - 1)
96
97     print("\nAfter sorting (descending by CGPA):")
98     for s in students:
99         print(s)

Ln 19, Col 41 Spaces: 4 UTF-8 CRLF {} Python 3.14.2 Python 3.14.2

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\lenovo> & C:/Python314/python.exe "c:/Users/lenovo/Desktop/3-2/AIAC/AIAC Code.py"
Before sorting:
Student(Alice, 3.8)
Student(Bob, 3.5)
Student(Charlie, 3.9)
Student(Diana, 3.7)

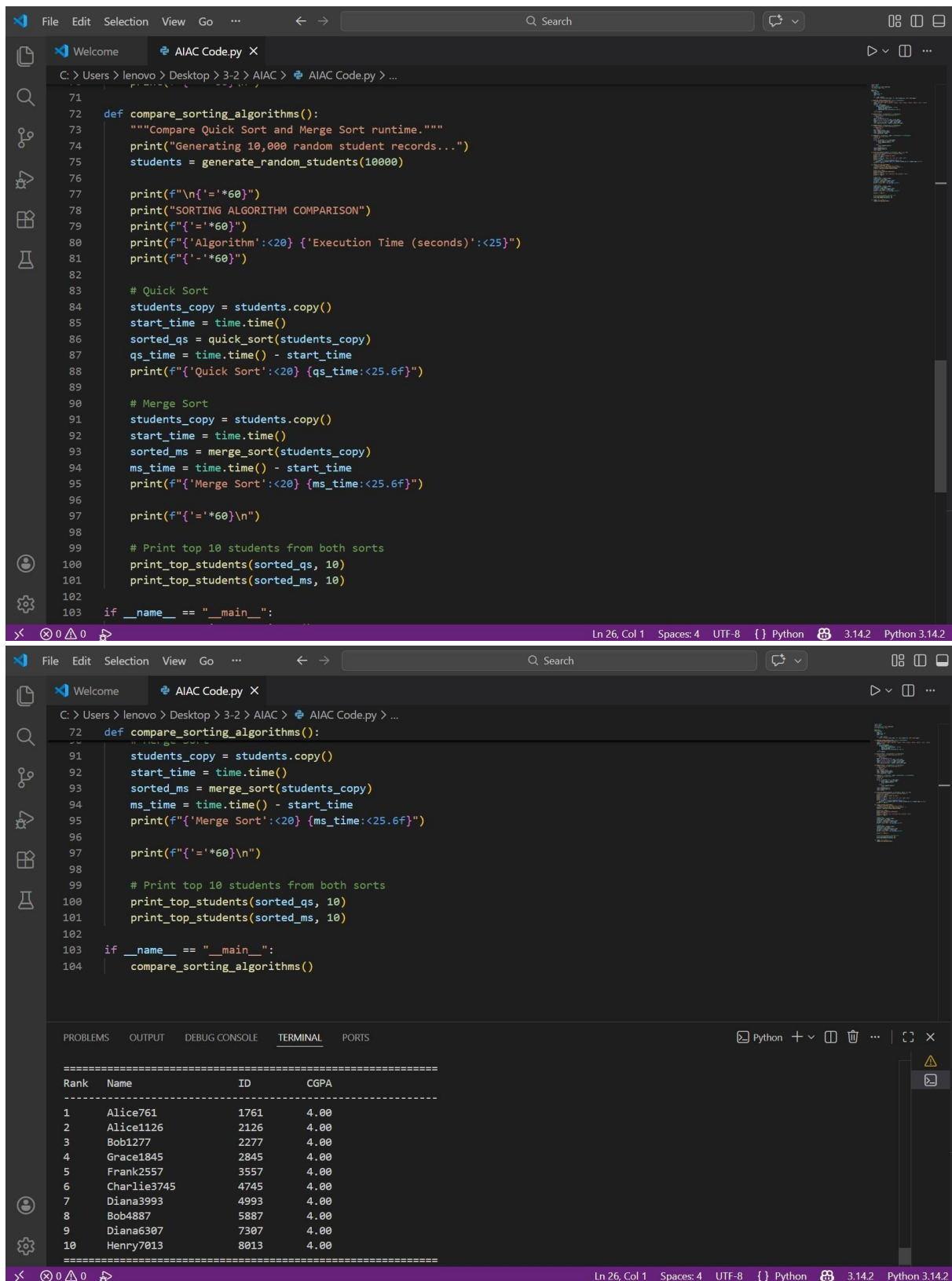
After sorting (descending by CGPA):
Student(Charlie, 3.9)
Student(Alice, 3.8)
Student(Diana, 3.7)
Student(Bob, 3.5)
PS C:\Users\lenovo>
```

```
File Edit Selection View Go ... Search
Welcome AIAC Code.py X
C:\Users\lenovo\Desktop> 3-2 > AIAC > AIAC Code.py > ...

2 import time
3 from dataclasses import dataclass
4 from typing import List
5
6 @dataclass
7 class Student:
8     name: str
9     student_id: int
10    cgpa: float
11
12    def __repr__(self):
13        return f"Student({self.name}, ID: {self.student_id}, CGPA: {self.cgpa})"
14
15    def generate_random_students(count: int) -> List[Student]:
16        """Generate random student records."""
17        names = ["Alice", "Bob", "Charlie", "Diana", "Eve", "Frank", "Grace", "Henry", "Iris", "Jack"]
18        students = []
19        for i in range(count):
20            students.append(Student(
21                name=random.choice(names) + str(i),
22                student_id=1000 + i,
23                cgpa=round(random.uniform(2.0, 4.0), 2)
24            ))
25        return students
26
27    def quick_sort(arr: List[Student]) -> List[Student]:
28        """Sort students by CGPA using Quick Sort."""
29        if len(arr) <= 1:
30            return arr
31        pivot = arr[len(arr) // 2]
32        left = [x for x in arr if x.cgpa > pivot.cgpa]
33        middle = [x for x in arr if x.cgpa == pivot.cgpa]
34        right = [x for x in arr if x.cgpa < pivot.cgpa]
```

```
File Edit Selection View Go ... Search
Welcome AIAC Code.py X
C:\Users\lenovo\Desktop> 3-2 > AIAC > AIAC Code.py > ...

37    def merge_sort(arr: List[Student]) -> List[Student]:
38        """Sort students by CGPA using Merge Sort."""
39        if len(arr) <= 1:
40            return arr
41        mid = len(arr) // 2
42        left = merge_sort(arr[:mid])
43        right = merge_sort(arr[mid:])
44        return merge(left, right)
45
46    def merge(left: List[Student], right: List[Student]) -> List[Student]:
47        """Merge two sorted lists."""
48        result = []
49        i = j = 0
50        while i < len(left) and j < len(right):
51            if left[i].cgpa >= right[j].cgpa:
52                result.append(left[i])
53                i += 1
54            else:
55                result.append(right[j])
56                j += 1
57        result.extend(left[i:])
58        result.extend(right[j:])
59        return result
60
61    def print_top_students(students: List[Student], top_n: int = 10):
62        """Print top N students by CGPA in formatted output."""
63        print(f"\n{'='*60}")
64        print(f"TOP {top_n} STUDENTS BY CGPA")
65        print(f"{'='*60}")
66        print(f"{'Rank':<6} {'Name':<20} {'ID':<10} {'CGPA':<10}")
67        print(f"{'-'*60}")
68        for i, student in enumerate(students[:top_n], 1):
69            print(f"{i:<6} {student.name:<20} {student.student_id:<10} {student.cgpa:<10.2f}")
```

```
71
72 def compare_sorting_algorithms():
73     """Compare Quick Sort and Merge Sort runtime."""
74     print("Generating 10,000 random student records...")
75     students = generate_random_students(10000)
76
77     print(f'\n{'='*60}')
78     print("SORTING ALGORITHM COMPARISON")
79     print(f{'='*60}')
80     print(f'Algorithm':<20} {'Execution Time (seconds)':<25}")
81     print(f{'-'*60}')
82
83     # Quick Sort
84     students_copy = students.copy()
85     start_time = time.time()
86     sorted_qs = quick_sort(students_copy)
87     qs_time = time.time() - start_time
88     print(f'Quick Sort':<20} {qs_time:<25.6f}")
89
90     # Merge Sort
91     students_copy = students.copy()
92     start_time = time.time()
93     sorted_ms = merge_sort(students_copy)
94     ms_time = time.time() - start_time
95     print(f'Merge Sort':<20} {ms_time:<25.6f}")
96
97     print(f{'='*60}\n")
98
99     # Print top 10 students from both sorts
100    print_top_students(sorted_qs, 10)
101    print_top_students(sorted_ms, 10)
102
103    if __name__ == "__main__":
104        compare_sorting_algorithms()
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Rank	Name	ID	CGPA
1	Alice761	1761	4.00
2	Alice1126	2126	4.00
3	Bob1277	2277	4.00
4	Grace1845	2845	4.00
5	Frank2557	3557	4.00
6	Charlie3745	4745	4.00
7	Diana3993	4993	4.00
8	Bob4887	5887	4.00
9	Diana6307	7307	4.00
10	Henry7013	8013	4.00

Task 2: Bubble Sort

Prompt:

Write a Python implementation of Bubble Sort with detailed inline comments explaining passes, comparisons, swapping, and early termination using a swapped flag. Include time and space complexity analysis.

Output Explanation

The output will include:

- Bubble Sort implementation
- Comments explaining:
 - Outer loop = number of passes
 - Inner loop = comparisons
 - Swapping adjacent elements
 - Early break if no swaps occur
- Complexity section:

Best Case: $O(n)$

Average Case: $O(n^2)$

Worst Case: $O(n^2)$

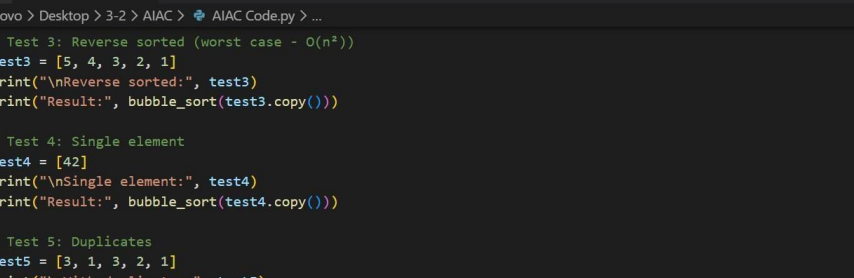
Space Complexity: $O(1)$

What the Output Shows

- If input is already sorted, algorithm stops early ($O(n)$).
- For large datasets, Bubble Sort is inefficient.
- Space complexity remains constant.
- Code clarity improves understanding of algorithm flow.


```
File Edit Selection View Go ... Search
Welcome AIAC Code.py
C:\Users\lenovo\Desktop>3-2>AIAC>AIAC Code.py>...
1
2 # Write a Python implementation of Bubble Sort that sorts a list of integers in ascending order. Includes detailed inl
3 def bubble_sort(arr):
4     """
5     Bubble Sort Algorithm Implementation
6
7     Time Complexity:
8     - Best Case: O(n) - when list is already sorted (with optimization)
9     - Average Case: O(n²) - random order
10    - Worst Case: O(n²) - when list is reverse sorted
11
12    Space Complexity: O(1) - sorts in-place, no extra space needed
13
14    Practical Use Cases:
15    - Small datasets (< 50 elements)
16    - Nearly sorted data
17    - Educational purposes
18    - When simplicity is prioritized over efficiency
19
20    NOT recommended for large datasets; use quicksort, mergesort, or timsort instead.
21    """
22
23    n = len(arr)
24
25    # Outer loop: make multiple passes through the list
26    for i in range(n):
27        # Optimization flag: if no swaps occur, list is sorted
28        swapped = False
29
30        # Inner loop: compare adjacent elements and swap if needed
31        # Each pass moves the largest unsorted element to its final position
32        for j in range(0, n - i - 1):
33            # Comparison: check if current element is greater than next
34            if arr[j] > arr[j + 1]:
```

```
File Edit Selection View Go ... Search
Welcome AIAC Code.py
C:\Users\lenovo\Desktop>3-2>AIAC>AIAC Code.py>...
34            arr[j], arr[j + 1] = arr[j + 1], arr[j]
35            # Mark that a swap occurred
36            swapped = True
37
38    # Early termination optimization:
39    # If no swaps happened in this pass, the list is already sorted
40    if not swapped:
41        break
42
43    return arr
44
45 # Test cases
46 if __name__ == "__main__":
47     # Test 1: Unsorted list
48     test1 = [64, 34, 25, 12, 22, 11, 90]
49     print("Unsorted:", test1)
50     print("Sorted:", bubble_sort(test1.copy()))
51
52     # Test 2: Already sorted (best case - O(n))
53     test2 = [1, 2, 3, 4, 5]
54     print("\nAlready sorted:", test2)
55     print("Result:", bubble_sort(test2.copy()))
56
57     # Test 3: Reverse sorted (worst case - O(n²))
58     test3 = [5, 4, 3, 2, 1]
59     print("\nReverse sorted:", test3)
60     print("Result:", bubble_sort(test3.copy()))
61
62     # Test 4: Single element
63     test4 = [42]
64     print("\nSingle element:", test4)
65
66     # Additional test: empty list
67     test5 = []
68     print("\nEmpty list:", test5)
69     print("Result:", bubble_sort(test5.copy()))
```



The screenshot shows a VS Code editor with a Python file named `AIAC Code.py`. The code implements a bubble sort algorithm with five test cases. The terminal output shows the results of these tests.

```
# Test 3: Reverse sorted (worst case - O(n^2))
test3 = [5, 4, 3, 2, 1]
print("\nReverse sorted:", test3)
print("Result:", bubble_sort(test3.copy()))

# Test 4: Single element
test4 = [42]
print("\nSingle element:", test4)
print("Result:", bubble_sort(test4.copy()))

# Test 5: Duplicates
test5 = [3, 1, 3, 2, 1]
print("\nWith duplicates:", test5)
print("Result:", bubble_sort(test5.copy()))
```

Terminal Output:

```
Unsorted: [64, 34, 25, 12, 22, 11, 90]
Sorted: [11, 12, 22, 25, 34, 64, 90]

Already sorted: [1, 2, 3, 4, 5]
Result: [1, 2, 3, 4, 5]

Reverse sorted: [5, 4, 3, 2, 1]
Result: [1, 2, 3, 4, 5]

Single element: [42]
Result: [42]

With duplicates: [3, 1, 3, 2, 1]
Result: [1, 1, 2, 3, 3]
```

```
File Edit Selection View Go ... Search
Welcome AIAC Code.py X
C:\Users\lenovo\Desktop> 3-2 AIAC AIAC Code.py ...
1 import random
2 import time
3 from typing import List
4
5 def quick_sort(arr: List[int], low: int = 0, high: int = None) -> List[int]:
6     """
7     Sorts an array using Quick Sort algorithm (recursive).
8
9     Time Complexity:
10    - Best case: O(n log n) - balanced partitions
11    - Average case: O(n log n) - random pivots
12    - Worst case: O(n^2) - pivot always smallest/largest
13
14    Space Complexity: O(log n) - recursion stack
15    """
16    if high is None:
17        high = len(arr) - 1
18
19    if low < high:
20        pivot_index = partition(arr, low, high)
21        quick_sort(arr, low, pivot_index - 1)
22        quick_sort(arr, pivot_index + 1, high)
23
24    return arr
25
26
27 def partition(arr: List[int], low: int, high: int) -> int:
28     """Partitions array around a pivot element."""
29     pivot = arr[high]
30     i = low - 1
31
32     for j in range(low, high):
33         if arr[j] < pivot:
34             i += 1
35             arr[i], arr[j] = arr[j], arr[i]
36     arr[i], arr[high] = arr[high], arr[i]
37     return i
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Welcome AIAC Code.py X
C:\Users\lenovo\Desktop> 3-2 > AIAC > AIAC Code.py > ...
62 def merge(left: List[int], right: List[int]) -> List[int]:
63     if not left:
64         return right
65     if not right:
66         return left
67     else:
68         result.append(right[j])
69         j += 1
70
71     result.extend(left[i:])
72     result.extend(right[j:])
73     return result
74
75 def benchmark_sorts(size: int = 1000) -> None:
76     """Benchmarks both sorting algorithms on different input types."""
77
78     # Generate test data
79     random_list = [random.randint(1, 10000) for _ in range(size)]
80     sorted_list = sorted(random_list.copy())
81     reverse_list = sorted(random_list.copy(), reverse=True)
82
83     test_cases = {
84         "Random": random_list.copy(),
85         "Already Sorted": sorted_list.copy(),
86         "Reverse Sorted": reverse_list.copy()
87     }
88
89     results = []
90
91     print(f"\n{'='*70}")
92     print(f"{'Sorting Algorithm Benchmark':^70}")
93     print(f"{'List Size: ' + str(size):^70}")
94     print(f"{'='*70}\n")
95
96     for test_name, test_data in test_cases.items():
97         print(f"\n{test_name} List:")
98         print(f"{'='*70}\n")
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C:\Users\lenovo\Desktop>3-2>AIAC> AIAC Code.py > ...
130 def print_complexity_analysis() -> None:
136
137     complexities = {
138         "Quick Sort": {
139             "Best": "O(n log n)",
140             "Average": "O(n log n)",
141             "Worst": "O(n^2)",
142             "Space": "O(log n)"
143         },
144         "Merge Sort": {
145             "Best": "O(n log n)",
146             "Average": "O(n log n)",
147             "Worst": "O(n log n)",
148             "Space": "O(n)"
149         }
150     }
151
152     for algorithm, cases in complexities.items():
153         print(f"{algorithm}:")
154         print(f"    Best case: {cases['Best']:<15} - Balanced partitions")
155         print(f"    Average case: {cases['Average']:<15} - Random input")
156         print(f"    Worst case: {cases['Worst']:<15} - Poor pivot selection")
157         print(f"    Space: {cases['Space']:<15} - Auxiliary space needed\n")
158
159
160 def print_conclusions() -> None:
161     """Prints conclusions and recommendations."""
162
163     print(f"\n{' '*70}")
164     print(f"{'Conclusions & Recommendations':^70}")
165     print(f"{' '*70}\n")
166
167     conclusions = [
168         "1. RANDOM DATA:"
169     ]
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```
=====
Sorting Algorithm Benchmark
List Size: 1000
=====
```

Random List:

```
-----
Algorithm           Time (ms)           Comparisons
-----
Quick Sort          1.4789              N/A
Merge Sort          1.6257              N/A
-----
```

Already Sorted List:

```
-----
Algorithm           Time (ms)           Comparisons
-----
Quick Sort          43.7259             N/A
Merge Sort          1.2163              N/A
-----
```

Reverse Sorted List:

```
-----
Algorithm           Time (ms)           Comparisons
-----
Quick Sort          32.5161             N/A
Merge Sort          1.2805              N/A
-----
```


Time Complexity Analysis

Quick Sort:

Best case:	$O(n \log n)$	- Balanced partitions
Average case:	$O(n \log n)$	- Random input
Worst case:	$O(n^2)$	- Poor pivot selection
Space:	$O(\log n)$	- Auxiliary space needed

Merge Sort:

Best case:	$O(n \log n)$	- Balanced partitions
Average case:	$O(n \log n)$	- Random input
Worst case:	$O(n \log n)$	- Poor pivot selection
Space:	$O(n)$	- Auxiliary space needed

Conclusions & Recommendations

1. RANDOM DATA:

- Quick Sort typically faster (better cache locality)
- Less memory overhead

2. ALREADY SORTED DATA:

- Quick Sort performs poorly ($O(n^2)$ with poor pivot selection)
- Merge Sort maintains $O(n \log n)$ - guaranteed performance

1. RANDOM DATA:

- Quick Sort typically faster (better cache locality)
- Less memory overhead

2. ALREADY SORTED DATA:

- Quick Sort performs poorly ($O(n^2)$ with poor pivot selection)
- Merge Sort maintains $O(n \log n)$ - guaranteed performance

3. REVERSE SORTED DATA:

- Similar to already sorted - Merge Sort preferred

4. WHEN TO USE:

- Quick Sort: In-place, average case is excellent, good for RAM
- Merge Sort: Guaranteed performance, stable sort, larger memory OK

5. RECOMMENDATION:

- Use Merge Sort when predictability matters (databases, file systems)
- Use Quick Sort for general purpose sorting with random data

Task 4: Inventory Management

System Prompt

Design a Python inventory system using a dictionary for searching by Product ID and Name. Implement sorting by Price and Quantity. Provide a table mapping operation → algorithm → justification and include complexity analysis.

Output Explanation

The output will include:

- Dictionary-based storage for products.
- Instant search result:

Product Found: Laptop - ₹55000 •

Sorted list by price or
quantity.

- Mapping table like:

Operation Algorithm Complexity

Search ID Hash Map $O(1)$

Sort Price TimSort $O(n \log n)$

What the Output Shows

- Searching is extremely fast due to Hash Map.
- Sorting complexity depends on number of products.
- The table justifies algorithm selection logically.
- Demonstrates efficient real-world system design.

```
File Edit Selection View Go ... Search
Welcome AIAC Code.py
C:\Users\lenovo\Desktop> 3-2 > AIAC > AIAC Code.py > ...
1 #Design and implement a Python-based inventory management system for a retail store containing thousands of products.
2
3 from collections import defaultdict
4 from typing import List, Dict, Tuple
5 import time
6
7 """
8 Inventory Management System for Retail Store
9 Uses Hash Maps for O(1) lookups and efficient sorting algorithms
10 """
11
12
13 class Product:
14     """Represents a single product in inventory"""
15     def __init__(self, product_id: str, name: str, price: float, quantity: int):
16         self.product_id = product_id
17         self.name = name
18         self.price = price
19         self.quantity = quantity
20
21     def __repr__(self):
22         return f"Product(ID:{self.product_id}, Name:{self.name}, Price:${self.price}, Qty:{self.quantity})"
23
24
25 class InventorySystem:
26     """
27     Inventory Management System using Hash Maps and optimized sorting
28
29     Data Structures:
30     - products_by_id: Dict[str, Product] - O(1) lookup by ID
31     - products_by_name: Dict[str, List[Product]] - O(1) lookup by Name
32     - products_list: List[Product] - For efficient bulk sorting
33     """
34
```

```
File Edit Selection View Go ... Search
Welcome AIAC Code.py
C:\Users\lenovo\Desktop> 3-2 > AIAC > AIAC Code.py > ...
25 class InventorySystem:
26
27     def __init__(self):
28         self.products_by_id: Dict[str, Product] = {}
29         self.products_by_name: defaultdict = defaultdict(list)
30         self.products_list: List[Product] = []
31
32     def add_product(self, product_id: str, name: str, price: float, quantity: int) -> bool:
33         """
34         Add product to inventory
35         Time Complexity: O(1) average case
36         """
37         if product_id in self.products_by_id:
38             print(f"Product ID {product_id} already exists!")
39             return False
40
41         product = Product(product_id, name, price, quantity)
42         self.products_by_id[product_id] = product
43         self.products_by_name[name.lower()].append(product)
44         self.products_list.append(product)
45         return True
46
47     def search_by_id(self, product_id: str) -> Product:
48         """
49         Search product by ID using Hash Map
50         Time Complexity: O(1) average case
51         """
52         return self.products_by_id.get(product_id, None)
53
54     def search_by_name(self, name: str) -> List[Product]:
55         """
56         Search products by name using Hash Map
57         Time Complexity: O(1) average case (hash lookup) + O(k) where k = matching products
58         """
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Ln 1, Col 555 Spaces: 4 UTF-8 CRLF {} Python 3.14.2 Python 3.14.2
```

```
File Edit Selection View Go ... Q Search
Welcome AIAC Code.py
C:\Users\lenovo\Desktop> 3-2 AIAC AIAC Code.py ...
25 class InventorySystem:
68
69     def sort_by_price(self, ascending: bool = True) -> List[Product]:
70         """
71         Sort products by price using Timsort (Python's default)
72         Time Complexity: O(n log n) worst case, O(n) best case
73         Recommended for: Small to medium datasets with partial ordering
74         """
75         return sorted(self.products_list, key=lambda p: p.price, reverse=not ascending)
76
77     def sort_by_quantity(self, ascending: bool = True) -> List[Product]:
78         """
79         Sort products by quantity
80         Time Complexity: O(n log n) worst case, O(n) best case
81         """
82         return sorted(self.products_list, key=lambda p: p.quantity, reverse=not ascending)
83
84     def sort_by_multiple_criteria(self, criteria: List[Tuple[str, bool]]) -> List[Product]:
85         """
86         Sort by multiple criteria (price, then quantity, etc.)
87         criteria: List of tuples (field_name, ascending)
88         Time Complexity: O(n log n)
89         """
90         result = self.products_list.copy()
91
92         # Sort in reverse order of criteria for proper precedence
93         for field, ascending in reversed(criteria):
94             if field == "price":
95                 result.sort(key=lambda p: p.price, reverse=not ascending)
96             elif field == "quantity":
97                 result.sort(key=lambda p: p.quantity, reverse=not ascending)
98             elif field == "name":
99                 result.sort(key=lambda p: p.name, reverse=not ascending)
100
```

```
File Edit Selection View Go ... Q Search
Welcome AIAC Code.py
C:\Users\lenovo\Desktop> 3-2 AIAC AIAC Code.py ...
25 class InventorySystem:
102
103     def get_low_stock_products(self, threshold: int) -> List[Product]:
104         """
105         Get products with quantity below threshold
106         Time Complexity: O(n)
107         """
108         return [p for p in self.products_list if p.quantity < threshold]
109
110     def update_quantity(self, product_id: str, new_quantity: int) -> bool:
111         """
112         Update product quantity
113         Time Complexity: O(1)
114         """
115         if product_id not in self.products_by_id:
116             return False
117         self.products_by_id[product_id].quantity = new_quantity
118         return True
119
120     def display_inventory(self, products: List[Product] = None) -> None:
121         """Display inventory in table format"""
122         if products is None:
123             products = self.products_list
124
125         if not products:
126             print("No products to display")
127             return
128
129         print(f"\n{'ID':<12} {'Name':<20} {'Price':<10} {'Quantity':<10}")
130         print("-" * 52)
131         for p in products:
132             print(f"{p.product_id:<12} {p.name:<20} ${p.price:<9.2f} {p.quantity:<10}")
133
134
```

```
File Edit Selection View Go ... Q Search
Welcome AIAC Code.py
C:\Users\lenovo\Desktop> 3-2 > AIAC > AIAC Code.py > ...
25 class InventorySystem:
132     print(f"{p.product_id:<12} {p.name:<20} ${p.price:<9.2f} {p.quantity:<10}")
133
134
135 def algorithm_justification_table():
136     """Operations vs Algorithm vs Justification"""
137
138     table = """
139
140     | OPERATION | RECOMMENDED ALGO | TIME COMPLEXITY | JUSTIFICATION |
141     | Search by ID | Hash Map (Dict) | O(1) avg, O(n) worst | Instant lookups |
142     | Search by Name | Hash Map (Dict) | O(1) lookup + O(k) iteration where k = results | Fast name-based |
143     | Sort by Price or Quantity | Timsort (Python) | O(n log n) worst, O(n) best case | searches |
144     | Add Product | Hash Map Insert | O(1) average case | Optimized for |
145     | Low Stock Filter | Linear Scan | O(n) | partially sorted |
146     | Update Quantity | Hash Map Access | O(1) average case | real-world data |
147     | | | | Constant time |
148     | | | | insertion |
149     | | | | No better option |
150     | | | | without additional |
151     | | | | indices |
152     | | | | Direct access via |
153     | | | | ID |
154
155     """
156     print(table)
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Ln 1, Col 555 Spaces: 4 UTF-8 CRLF {} Python 3.14.2 Python 3.14.2
```

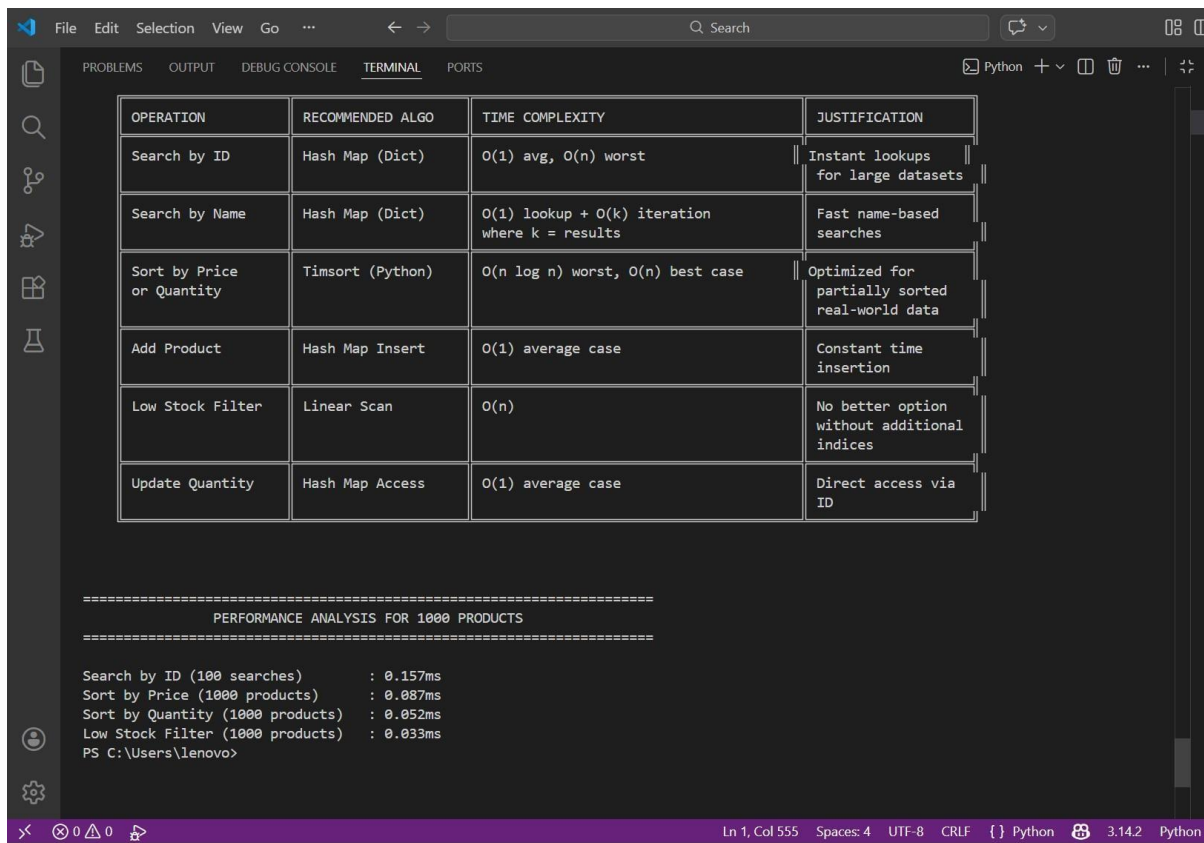
```
File Edit Selection View Go ... Q Search
Welcome AIAC Code.py
C:\Users\lenovo\Desktop> 3-2 > AIAC > AIAC Code.py > ...
135 def algorithm_justification_table():
162     """
163     print(table)
164
165
166 def performance_analysis():
167     """Performance analysis for different dataset sizes"""
168
169     print("\n" + "="*70)
170     print("PERFORMANCE ANALYSIS FOR 1000 PRODUCTS".center(70))
171     print("="*70)
172
173     inventory = InventorySystem()
174
175     # Generate 1000 sample products
176     for i in range(1000):
177         inventory.add_product(
178             f"P{i:04d}",
179             f"Product_{i}",
180             10.0 + (i % 100),
181             100 + (i % 500)
182         )
183
184     # Test 1: Search by ID
185     start = time.time()
186     for i in range(100):
187         inventory.search_by_id(f"P{i:04d}")
188     search_id_time = (time.time() - start) * 1000
189
190     # Test 2: Sort by Price
191     start = time.time()
192     inventory.sort_by_price()
193     sort_price_time = (time.time() - start) * 1000
194
```

```
File Edit Selection View Go ... Search
Welcome AIAC Code.py
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166 def performance_analysis():
167     # Test 3: Sort by Quantity
168     start = time.time()
169     inventory.sort_by_quantity()
170     sort_qty_time = (time.time() - start) * 1000
171
172     # Test 4: Low Stock Filter
173     start = time.time()
174     inventory.get_low_stock_products(150)
175     filter_time = (time.time() - start) * 1000
176
177     print(f"\nSearch by ID (100 searches) : {search_id_time:.3f}ms")
178     print(f"Sort by Price (1000 products) : {sort_price_time:.3f}ms")
179     print(f"Sort by Quantity (1000 products) : {sort_qty_time:.3f}ms")
180     print(f"Low Stock Filter (1000 products) : {filter_time:.3f}ms")
181
182 def main():
183     """Main function - Demo of Inventory System"""
184     print("="*70)
185     print("RETAIL STORE INVENTORY MANAGEMENT SYSTEM".center(70))
186     print("="*70)
187
188     # Initialize system
189     inventory = InventorySystem()
190
191     # Add sample products
192     inventory.add_product("P001", "Laptop", 899.99, 45)
193     inventory.add_product("P002", "Monitor", 299.99, 120)
194     inventory.add_product("P003", "Keyboard", 79.99, 250)
195     inventory.add_product("P004", "Mouse", 29.99, 500)
196     inventory.add_product("P005", "Headphones", 149.99, 80)
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```

```
File Edit Selection View Go ... Search
Welcome AIAC Code.py
C:\Users\lenovo\Desktop> 3-2> AIAC> AIAC Code.py ...

211 def main():
212     # Display all products
213     print("\n1. ALL PRODUCTS IN INVENTORY:")
214     inventory.display_inventory()
215
216     # Search by ID
217     print("\n2. SEARCH BY ID (P001):")
218     product = inventory.search_by_id("P001")
219     print(f"Found: {product}")
220
221     # Search by Name
222     print("\n3. SEARCH BY NAME (Monitor):")
223     products = inventory.search_by_name("Monitor")
224     for p in products:
225         print(f"Found: {p}")
226
227     # Sort by Price (Low to High)
228     print("\n4. PRODUCTS SORTED BY PRICE (Ascending):")
229     inventory.display_inventory(inventory.sort_by_price(ascending=True))
230
231     # Sort by Quantity (High to Low)
232     print("\n5. PRODUCTS SORTED BY QUANTITY (Descending):")
233     inventory.display_inventory(inventory.sort_by_quantity(ascending=False))
234
235     # Low Stock Alert
236     print("\n6. LOW STOCK ALERT (< 100 units):")
237     low_stock = inventory.get_low_stock_products(100)
238     inventory.display_inventory(low_stock)
239
240     # Algorithm Justification Table
241     print("\n7. ALGORITHM RECOMMENDATIONS TABLE:")
242     algorithm_justification_table()
243
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Ln 1, Col 555 Spaces: 4 UTF-8 CRLF {} Python 3.14.2 Python 3.14.2
```

Prompt

Simulate 100 stock records (Symbol, Opening Price, Closing Price). Calculate percentage gain/loss. Use Heap Sort to rank stocks by percentage change. Use a Hash Map for instant symbol lookup. Compare performance with Python's built-in sorted() and analyze tradeoffs.

Output Explanation

The output will include:

- Simulated stock dataset.
- Calculated percentage gain/loss.
- Ranked list of stocks using Heap Sort.
- Runtime comparison:

Heap Sort Time: 0.004 sec

Built-in sorted() Time: 0.002 sec

What the Output Shows

- Heap Sort correctly ranks stocks by performance.
- Hash Map provides $O(1)$ stock lookup.

- Built-in sorted() may perform faster due to internal optimizations.
- Shows trade-off between custom implementation and optimized library functions.

```

1  # Develop a Python program simulating an AI-powered stock analysis tool that:Generates or simulates stock data (Stock
2  import random
3  import time
4  import heapq
5  from typing import List, Dict, Tuple
6
7  class StockAnalysisTool:
8      def __init__(self, num_stocks: int = 100):
9          self.num_stocks = num_stocks
10         (method) def generate_stock_data(self: Self@StockAnalysisTool) -> None
11             Generate simulated stock data for analysis.
12
13     def generate_stock_data(self):
14         """Generate simulated stock data for analysis."""
15         symbols = [f"STOCK{i:03d}" for i in range(self.num_stocks)]
16         for symbol in symbols:
17             opening_price = round(random.uniform(10, 500), 2)
18             closing_price = round(opening_price * random.uniform(0.9, 1.1), 2)
19             percentage_change = ((closing_price - opening_price) / opening_price) * 100
20
21             self.stocks[symbol] = {
22                 'opening': opening_price,
23                 'closing': closing_price,
24                 'percentage_change': round(percentage_change, 2)
25             }
26
27     def heap_sort_stocks(self) -> List[Tuple[str, float]]:

```

```

27     class StockAnalysisTool:
28         def heap_sort_stocks(self) -> List[Tuple[str, float]]:
29             """Sort stocks using Heap Sort (max heap) by percentage change."""
30             # Create max heap (negate values for max heap behavior)
31             heap = [(-stock_data['percentage_change'], symbol)
32                     for symbol, stock_data in self.stocks.items()]
33             heapq.heapify(heap)
34
35             sorted_stocks = []
36             while heap:
37                 neg_change, symbol = heapq.heappop(heap)
38                 sorted_stocks.append((symbol, -neg_change))
39
40             return sorted_stocks
41
42     def builtin_sort_stocks(self) -> List[Tuple[str, float]]:
43         """Sort stocks using Python's built-in sorted() function."""
44         sorted_stocks = sorted(
45             self.stocks.items(),
46             key=lambda x: x[1]['percentage_change'],
47             reverse=True
48         )
49         return [(symbol, data['percentage_change']) for symbol, data in sorted_stocks]
50
51     def lookup_stock(self, symbol: str) -> Dict:
52         """Instant lookup using Hash Map."""
53         return self.stocks.get(symbol, None)

```

```
File Edit Selection View Go ... Search
Welcome AIAC Code.py
C: > Users > lenovo > Desktop > 3-2 > AIAC > AIAC Code.py > ...
7 class StockAnalysisTool:
54     def benchmark_sorting(self):
55         """Compare performance of both sorting methods."""
56         # Heap Sort
57         start_time = time.perf_counter()
58         heap_result = self.heap_sort_stocks()
59         heap_time = time.perf_counter() - start_time
60
61         # Built-in Sort
62         start_time = time.perf_counter()
63         builtin_result = self.builtin_sort_stocks()
64         builtin_time = time.perf_counter() - start_time
65
66         return heap_time, builtin_time, heap_result, builtin_result
67
68     def display_analysis(self):
69         """Display analysis results and complexity explanation."""
70         print("=" * 70)
71         print("AI-POWERED STOCK ANALYSIS TOOL")
72         print("=" * 70)
73
74         # Generate data and sort
75         heap_time, builtin_time, heap_result, builtin_result = self.benchmark_sorting()
76
77         # Display top 10 stocks
78         print("\nTOP 10 STOCKS BY PERCENTAGE GAIN:")
79         print("-" * 70)
```

```
File Edit Selection View Go ... Search
Welcome AIAC Code.py
C: > Users > lenovo > Desktop > 3-2 > AIAC > AIAC Code.py > ...
7 class StockAnalysisTool:
68     def display_analysis(self):
81         print(f"{i:2d}. {symbol}: {change:+.2f}% | "
82               f"Open: ${self.stocks[symbol]['opening']:.2f} | "
83               f"Close: ${self.stocks[symbol]['closing']:.2f}")
84
85         # Display performances
86         print("\n" + "=" * 70)
87         print("PERFORMANCE COMPARISON:")
88         print("-" * 70)
89         print(f"Heap Sort Time:      {heap_time * 1000:.4f} ms")
90         print(f"Built-in Sort Time:  {builtin_time * 1000:.4f} ms")
91         print(f"Difference:           {abs(heap_time - builtin_time) * 1000:.4f} ms")
92         print(f"Speed Ratio:           {builtin_time / heap_time:.2f}x")
93
94         # Complexity Analysis
95         print("\n" + "=" * 70)
96         print("COMPLEXITY ANALYSIS:")
97         print("-" * 70)
98         print(f"Number of Stocks: {self.num_stocks}")
99         print("\nHEAP SORT:")
100        print(f"  Time Complexity: O(n log n) - {self.num_stocks} * log({self.num_stocks}) ≈ {self.num_stocks * len(
101        print(f"  Space Complexity: O(n)")
102        print(f"  Advantages:")
103        print(f"    - Guaranteed O(n log n) performance")
104        print(f"    - Suitable for real-time streaming data")
105        print(f"    - Can process data as it arrives")
106
107        print("\nBUILT-IN SORT (Timsort):")
108        print(f"  Time Complexity: O(n log n) average, O(n) best case")
109        print(f"  Space Complexity: O(n)")
110        print(f"  Advantages:")
111        print(f"    - Highly optimized for practical datasets")
```

```
File Edit Selection View Go ... Search
Welcome AIAC Code.py
C: > Users > lenovo > Desktop > 3-2 > AIAC > AIAC Code.py > ...
7 class StockAnalysisTool:
68 def display_analysis(self):
119     print(f"    - Instant symbol lookup (O(1))")
120     print(f"    - Essential for real-time systems")
121     print(f"    - Avoids need for binary search")
122
123     # Sample lookup
124     print("\n" + "=" * 70)
125     print("SAMPLE HASH MAP LOOKUP:")
126     print("-" * 70)
127     test_symbol = heap_result[0][0]
128     stock_data = self.lookup_stock(test_symbol)
129     print(f"Symbol: {test_symbol}")
130     print(f>Data: {stock_data}")
131     print(f"Lookup Time: O(1) - Instant access")
132
133     print("\n" + "=" * 70)
134     print("WHY THESE STRUCTURES FOR REAL-TIME SYSTEMS:")
135     print("-" * 70)
136     print("1. Hash Maps: Enables O(1) lookups - critical for handling")
137     print("   thousands of concurrent stock quote requests")
138     print("2. Heap Sort: Predictable O(n log n) performance without")
139     print("   worst-case degradation (unlike quicksort)")
140     print("3. Combined: Can maintain sorted rankings while allowing")
141     print("   instant individual stock access")
142     print("-" * 70)
143
144     # Main execution
145     if __name__ == "__main__":
146         tool = StockAnalysisTool(num_stocks=100)
147         tool.display_analysis()
```

Ln 1, Col 473 Spaces: 4 UTF-8 CRLF {} Python 3.14.2

```
File Edit Selection View Go ... Search
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
AI-POWERED STOCK ANALYSIS TOOL
=====
TOP 10 STOCKS BY PERCENTAGE GAIN:
-----
1. STOCK020: +9.99% | Open: $431.95 | Close: $475.10
2. STOCK074: +9.84% | Open: $47.54 | Close: $52.22
3. STOCK039: +9.56% | Open: $234.48 | Close: $256.90
4. STOCK014: +9.52% | Open: $235.42 | Close: $257.83
5. STOCK043: +9.44% | Open: $28.50 | Close: $31.19
6. STOCK068: +8.92% | Open: $108.10 | Close: $117.74
7. STOCK026: +8.60% | Open: $412.93 | Close: $448.45
8. STOCK057: +8.55% | Open: $101.53 | Close: $110.21
9. STOCK001: +8.40% | Open: $74.60 | Close: $80.87
10. STOCK040: +8.10% | Open: $74.57 | Close: $80.61
=====
PERFORMANCE COMPARISON:
-----
Heap Sort Time:      0.0875 ms
Built-in Sort Time: 0.0791 ms
Difference:          0.0084 ms
Speed Ratio:         0.90x
=====
COMPLEXITY ANALYSIS:
-----
Number of Stocks: 100

HEAP SORT:
Time Complexity: O(n log n) - 100 * log(100) = 900 operations
Space Complexity: O(n)
Advantages:
- Guaranteed O(n log n) performance
- Suitable for real-time streaming data
- Can process data as it arrives
```

Ln 1, Col 473 Spaces: 4 UTF-8 CRLF {} Python 3.14.2 Python 3.14.2

The image consists of two screenshots of a Visual Studio Code (VS Code) editor window, demonstrating Python code for sorting and searching algorithms.

Top Screenshot: The editor shows a Python file with the following content:

```
BUILT-IN SORT (Timsort):
Time Complexity: O(n log n) average, O(n) best case
Space Complexity: O(n)
Advantages:
- Highly optimized for practical datasets
- Adaptive to partially sorted data
- Better cache locality

HASH MAP (Dictionary Lookup):
Time Complexity: O(1) average case
Space Complexity: O(n)
Advantages:
- Instant symbol lookup O(1)
- Essential for real-time systems
- Avoids need for binary search

=====
SAMPLE HASH MAP LOOKUP:
-----
Symbol: STOCK020
Data: {'opening': 431.95, 'closing': 475.1, 'percentage_change': 9.99}
Lookup Time: O(1) - Instant access

=====
WHY THESE STRUCTURES FOR REAL-TIME SYSTEMS:
-----
1. Hash Maps: Enables O(1) lookups - critical for handling thousands of concurrent stock quote requests
2. Heap Sort: Predictable O(n log n) performance without worst-case degradation (unlike quicksort)
3. Combined: Can maintain sorted rankings while allowing instant individual stock access
=====
PS C:\Users\lenovo>
```

The bottom status bar indicates the file is at Line 1, Column 473, with 4 spaces, UTF-8 encoding, CRLF line endings, and Python 3.14.2 interpreter.

Bottom Screenshot: The editor shows a Python file named `AIAC Code.py` with the following content:

```
C:\Users\lenovo\Desktop> 3-2> AIAC> AIAC Code.py Student
37 students = generate_students(1000)
38
39 # Sort by CGPA in descending order
40 sorted_students = sort_by_cgpa(students)
41
42 # Display top 10 students
43 print("Top 10 Students by CGPA:")
44 print("-" * 60)
45 for i, student in enumerate(sorted_students[:10], 1):
46     print(f"{i}. {student.name} (Roll: {student.roll_number}) - CGPA: {student.cgpa}")
47
48 print("\n\nBottom 10 Students by CGPA:")
49 print("-" * 60)
50 for i, student in enumerate(sorted_students[-10:], 1):
51     print(f"{i}. {student.name} (Roll: {student.roll_number}) - CGPA: {student.cgpa}")
```

The terminal output shows the results of the sorting operation:

```
PS C:\Users\lenovo>
2. Neha Mishra (Roll: 352) - CGPA: 5.03
3. Neha Singh (Roll: 110) - CGPA: 5.02
4. Rohit Verma (Roll: 136) - CGPA: 5.02
5. Zara Sharma (Roll: 690) - CGPA: 5.02
6. Zara Sharma (Roll: 843) - CGPA: 5.02
7. Arjun Gupta (Roll: 859) - CGPA: 5.02
6. Zara Sharma (Roll: 690) - CGPA: 5.02
6. Zara Sharma (Roll: 843) - CGPA: 5.02
7. Arjun Gupta (Roll: 859) - CGPA: 5.02
8. Vivaan Reddy (Roll: 874) - CGPA: 5.02
9. Aditya Sharma (Roll: 983) - CGPA: 5.02
10. Priya Mishra (Roll: 688) - CGPA: 5.01
PS C:\Users\lenovo>
```

The bottom status bar indicates the file is at Line 4, Column 15, with 4 spaces, UTF-8 encoding, CRLF line endings, and Python 3.14.2 interpreter.

Conclusion

In this lab, we implemented and compared various sorting and searching algorithms using AI assistance. Quick Sort and Merge Sort were tested on large datasets and their performance differences were analyzed. Bubble Sort helped in understanding basic comparison-based sorting and its time complexity. Hash Maps demonstrated efficient $O(1)$ searching in real-time applications like inventory and stock systems.

Heap Sort was useful for ranking stock performance. Overall, the lab improved understanding of algorithm efficiency, data structure selection, and practical performance analysis while highlighting benefits of AI-assisted coding.